

Advanced Flying Robots – INDI Controller

Current Progress: Motion Planning + INDI on Crazyflie 2.1

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Motion Planning – INDI Controller

Motion planning pipeline

- Minimum-snap polynomial trajectories; two timing modes: Mode 1 (k_t timing), Mode 3 (time redistribution).
- Differential flatness maps the planned trajectory to the controller feedforward (thrust, attitude, $\dot{\omega}_d$).

INDI controller

- Incremental, sensor-based control law (Tal & Karaman 2020).
- Benchmarked on the CF2.1: 37% better tracking than geometric. Port to the thrust-upgraded / brushless drone is ongoing (Next Steps).

Planning Modes 1 And 3

Both modes share the same minimum-snap polynomial backbone and flatness-based attitude; they differ only in how the segment times T_i are chosen.

Mode	Timing idea	Segment-time rule
1	k_t -based allocation	$T_i = \max(d_i/v_{avg}, T_{\min}), \quad v_{avg} = 0.5 + 1.5\sqrt{k_t}$
3	Time redistribution	$T_i^{\text{tgt}} = T_{\text{tot}} \frac{J_i}{\sum_j J_j}$ (clamped to $T_{\min}$); then $T_i \leftarrow T_i + \frac{1}{2}(T_i^{\text{tgt}} - T_i)$, starting from Mode 1

T_i : time of segment i ; $d_i = \|p_{i+1} - p_i\|$: segment length; v_{avg} : average speed; k_t : speed / aggressiveness scalar; $T_{\min} = 0.15$ s: minimum segment time; J_i : snap cost of segment i ; T_{tot} : total trajectory time.

Core Objective: Minimum Snap As A QP

Segment polynomial and snap cost (what we minimize)

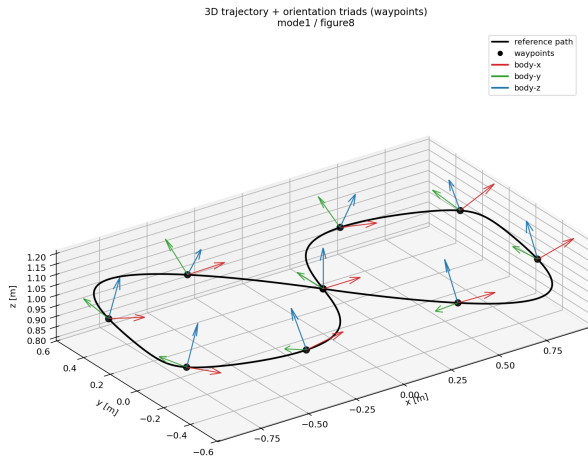
$$\sigma_i(\hat{t}) = \sum_{k=0}^8 c_{i,k} \hat{t}^k, \quad \hat{t} = \frac{\tau}{T_i} \in [0, 1], \quad J = \sum_i \frac{1}{T_i^7} \int_0^1 \left(\frac{d^4 \sigma_i}{d\hat{t}^4} \right)^2 d\hat{t}$$

Quadratic program

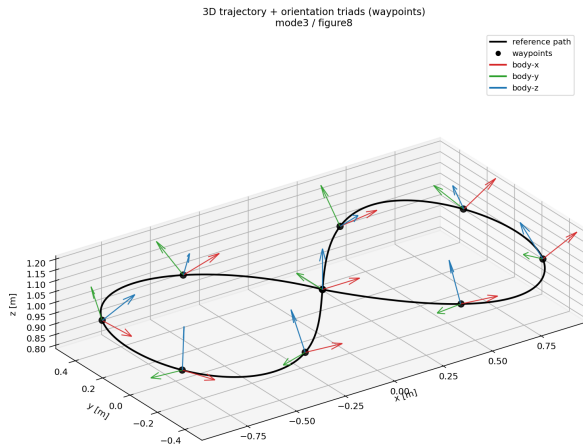
$$\min_{\mathbf{c}} \quad \frac{1}{2} \mathbf{c}^\top \mathbf{Q} \mathbf{c} + \mathbf{q}^\top \mathbf{c} \quad \text{s.t.} \quad \mathbf{A}_{eq} \mathbf{c} = \mathbf{b}_{eq}$$

Equality constraints only: waypoint interpolation $\sigma_i(0) = w_i$, $\sigma_i(1) = w_{i+1}$, and derivative continuity $\frac{1}{T_i^r} \sigma_i^{(r)}(1) = \frac{1}{T_{i+1}^r} \sigma_{i+1}^{(r)}(0)$ for $r = 1, \dots, 4$. Solved per axis; coupling re-enters through flatness. Thrust / rate / torque limits are checked *after* solving.

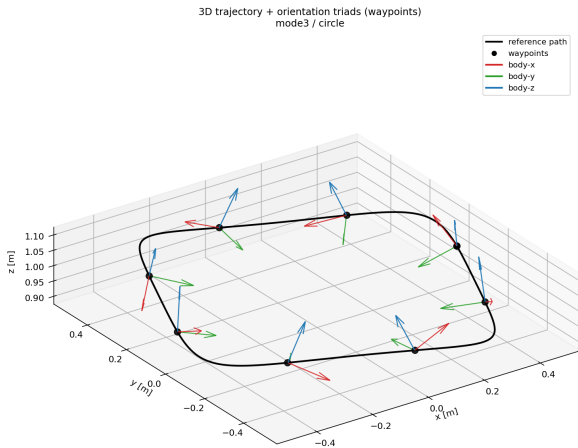
Mode 1 — Planned Figure-8 (3D Trajectory)



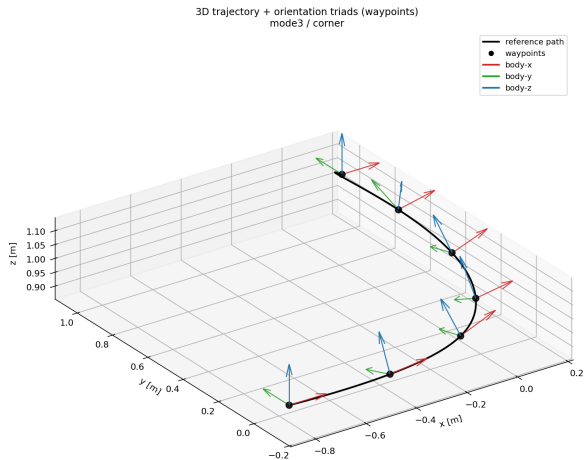
Mode 3 — Planned Figure-8 (3D Trajectory)



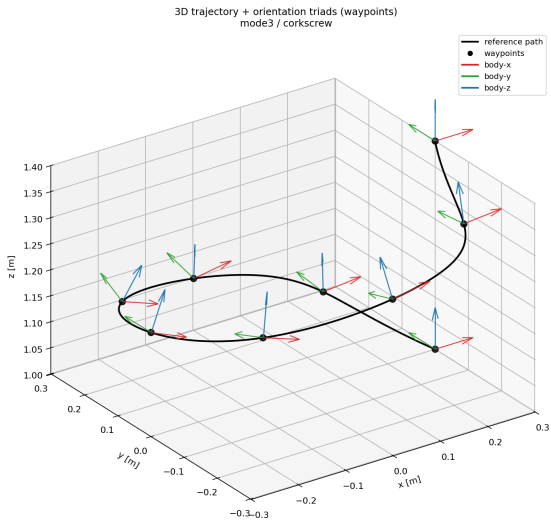
Mode 3 — Circle (3D Trajectory)



Mode 3 — Corner (3D Trajectory)

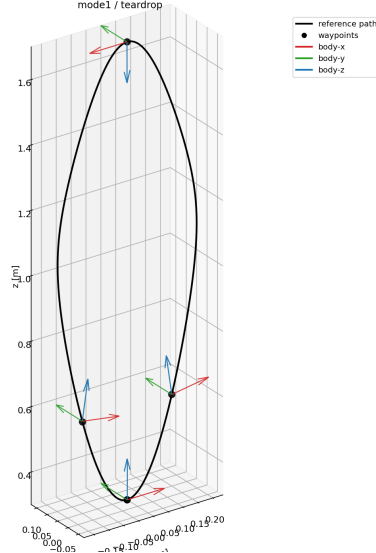


Mode 3 — Corkscrew (3D Trajectory)



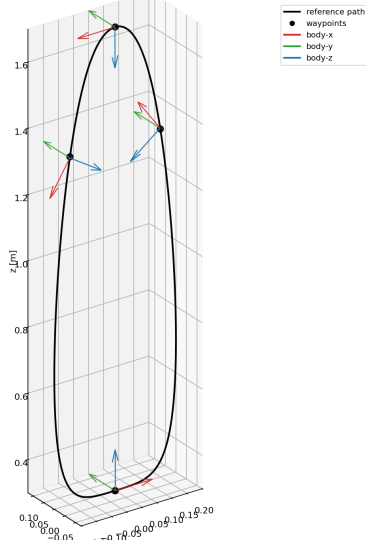
Teardrop — Round Bottom (3D Trajectory)

3D trajectory + orientation triads (waypoints)
model1 / teardrop



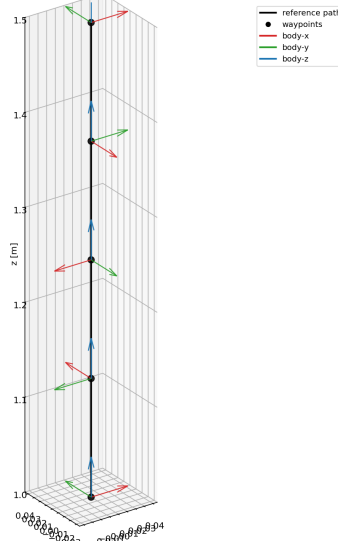
Teardrop — Round Top (3D Trajectory)

3D trajectory + orientation triads (waypoints)
model / teardrop_high

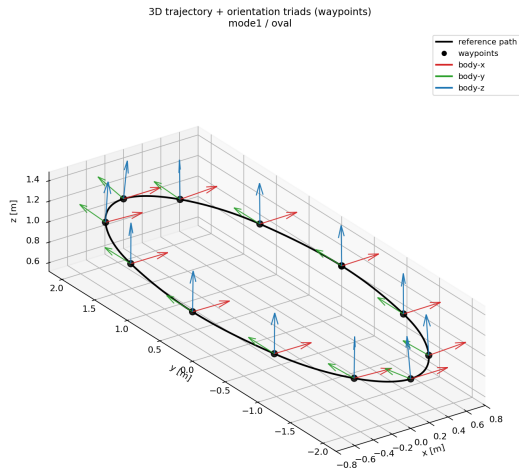


Mode 3 — Screw (3D Trajectory)

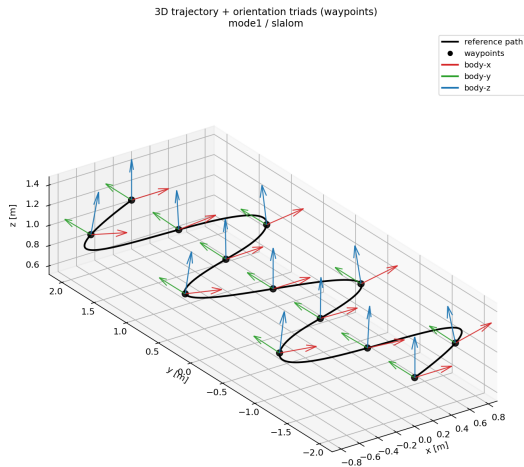
3D trajectory + orientation triads (waypoints)
mode3 / screw



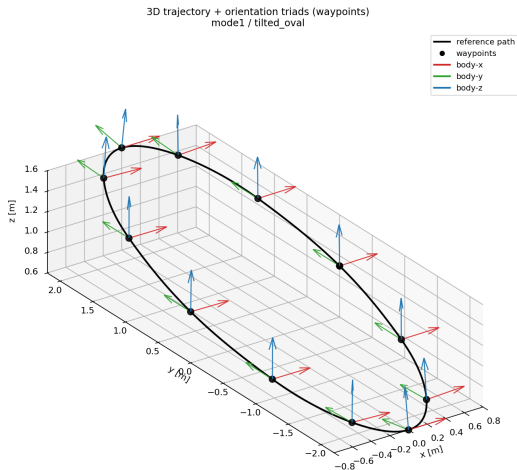
Racing Trajectory — Oval (3D Trajectory)



Racing Trajectory — Slalom (3D Trajectory)



Racing Trajectory — Tilted Oval (3D Trajectory)



Why INDI? Incremental, Sensor-Based Control

Geometric (model-based) attitude law

$$\tau = -K_R e_R - K_\omega e_\omega + \omega \times J\omega$$

where e_R : attitude error, e_ω : body-rate error, K_R, K_ω : torque gains, J : inertia.

Needs an accurate model (inertia, aerodynamics, actuator mapping). Model error $\rightarrow$ tracking error.

INDI idea — Incremental Nonlinear Dynamic Inversion

- Instead of the full model torque, measure the current angular acceleration + actuator torque and command only the *increment* needed to reach the desired angular acceleration.
- The measured $\dot{\omega}$ already contains the true dynamics + disturbances $\Rightarrow$ model error largely cancels.
- Applied to *both* loops: torque increment (attitude) and acceleration increment (position).

Reference: Tal & Karaman, “Accurate Tracking of Aggressive Quadrotor Trajectories”, 2020.

INDI Control Law — Algorithm

Outer loop — position (acceleration increment)

$$\begin{aligned}e_p &= p_d - p, & e_v &= v_d - v && \text{position / velocity error} \\a_{\text{meas}} &= R a_{\text{imu}} - g e_3, & a_{\text{model}} &= \frac{1}{m} \left(\sum_i k_{t,i} \text{rpm}_i^2 \right) z_B - g e_3 && \text{measured vs. modeled accel} \\f_d &= a_d + K_P e_p + K_V e_v + g e_3 + (a_{\text{meas}} - a_{\text{model}}) && \leftarrow \text{INDI term} \\T &= m f_d \cdot z_B && \text{collective thrust}\end{aligned}$$

Inner loop — attitude (torque increment)

$$\begin{aligned}e_R &= \frac{1}{2} (R_d^\top R - R^\top R_d)^\vee, & e_\omega &= \omega - \omega_d && \text{attitude / rate error} \\\dot{\omega}_{\text{ref}} &= \dot{\omega}_d - k_r e_R - k_w e_\omega && \text{desired angular accel} \\\tau &= \tau_{\text{cur}} + J(\dot{\omega}_{\text{ref}} - \dot{\omega}_{\text{meas}}) && \leftarrow \text{INDI term}\end{aligned}$$

Subscript d : trajectory reference; z_B : body z-axis; a_{imu} : accelerometer; K_P, K_V : position gains; $k_r [\text{s}^{-2}], k_w [\text{s}^{-1}]$: INDI inner gains ($\neq$ geometric torque gains). $\dot{\omega}_{\text{ref}}$ and $\dot{\omega}_{\text{meas}}$ share one Butterworth. Geometric mode = both INDI terms off.

INDI: Angular Acceleration And Control Effectiveness

Measured angular acceleration $\dot{\omega}_{\text{meas}}$ (the critical signal)

$$\dot{\omega}_{\text{meas}} = \text{LP} \left(\frac{\omega_k - \omega_{k-1}}{\Delta t} \right), \quad \text{2nd-order Butterworth, cutoff } f_{bw}$$

ω_k : current gyro sample, Δt : control period, f_{bw} : filter cutoff. Trade-off: lower $f_{bw} \rightarrow$ less noise but more phase lag; higher $\rightarrow$ faster but noisier.

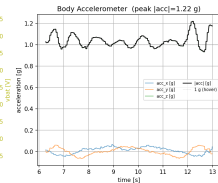
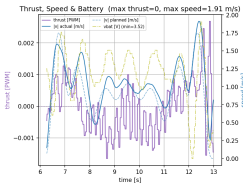
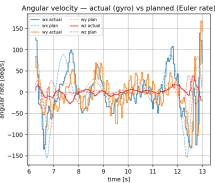
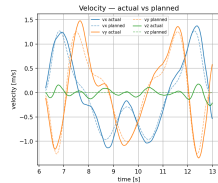
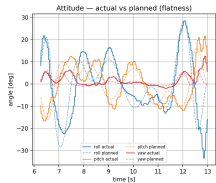
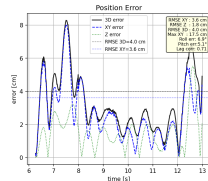
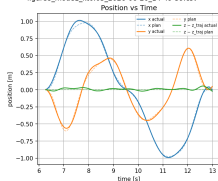
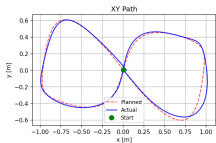
Current torque τ_{cur} from motor speeds

$$f_i = k_{t,i} \text{rpm}_i^2, \quad \tau_{\text{cur}} = \begin{bmatrix} \ell (f_3 + f_4 - f_1 - f_2) \\ \ell (f_2 + f_3 - f_1 - f_4) \\ c_{Q/T} (f_2 + f_4 - f_1 - f_3) \end{bmatrix}$$

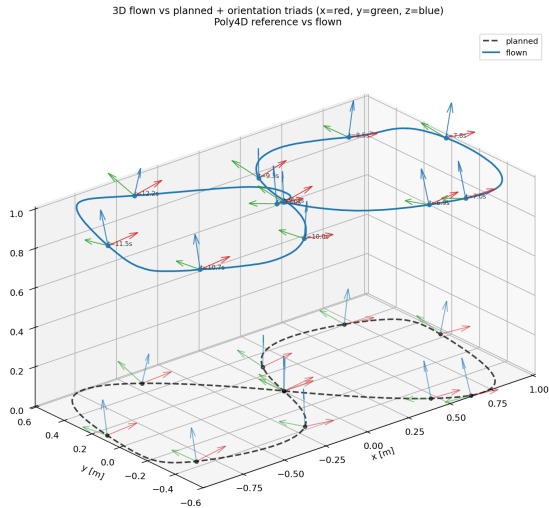
f_i : thrust of motor i , $k_{t,i}$: per-motor thrust constant, rpm_i : motor speed (RPM deck), ℓ : projected arm length, $c_{Q/T}$: torque-to-thrust ratio. τ_{cur} and $\dot{\omega}_{\text{meas}}$ feed the inner-loop increment.

INDI Flight: Best Figure-8 — Full Analysis

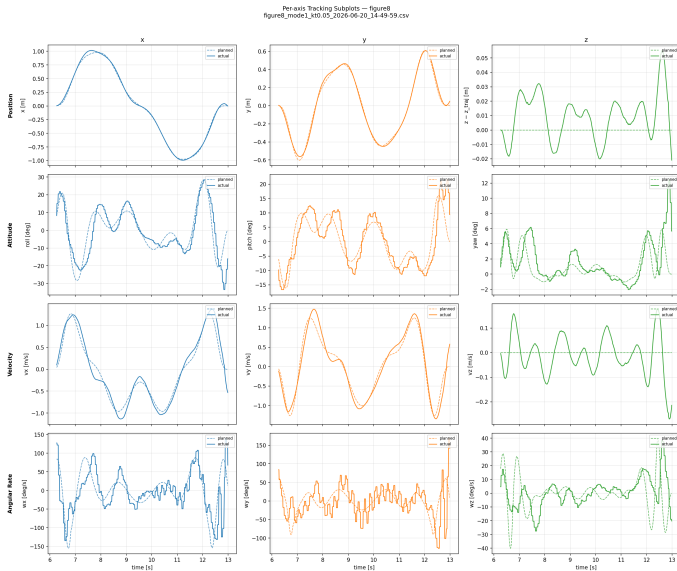
Trajectory Analysis — figure8 (SPEED_SCALE=1.0, XY_SCALE=1.0)
figure8_model1_kt05_2026-06-20_14-49-59.csv



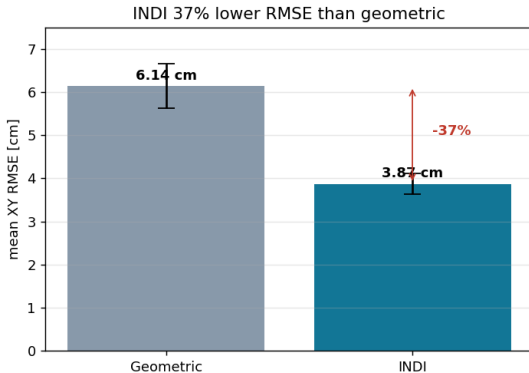
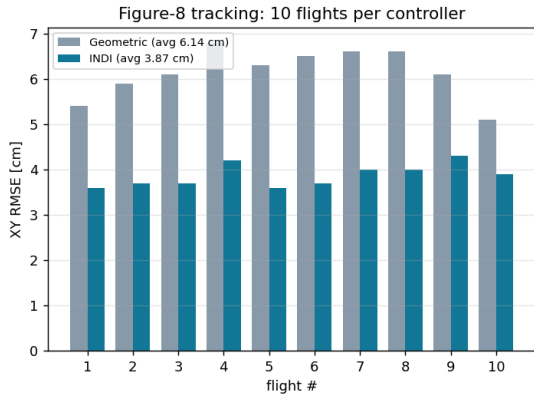
INDI Flight: Best Figure-8 — 3D Orientation



INDI Flight: Best Figure-8 — Per-Axis Tracking



Geometric vs. INDI — 10 Flights Each



INDI Speed Sweep — Figure-8 Aggressiveness Limit

k_t	Lap time [s]	XY RMSE	Status
0.01	8.84	2.8 cm	stable
0.02	8.07	3.4 cm	stable
0.03	7.57	4.7 cm	stable
0.04	7.19	4.0 cm	stable
0.05	6.88	5.4 cm	stable — best tracking
0.10	5.90	11.3 cm	stable — degrading
0.15	5.32	16.0 cm	stable — speed limit
0.20	4.91	—	crash — genuine instability

- ① INDI on the thrust-upgraded drone
- ② INDI on the brushless platform
- ③ Maximize figure-8 speed while stable
- ④ Fly more different trajectories